

Question 1 | Correct | Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,3]

the subsequence is [-1,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 int main() {
3     int n;
4     scanf("%d", &n);
5     int a[n], dp[n];
6     for (int i = 0; i < n; i++) {
7         scanf("%d", &a[i]);
8         dp[i] = 1;
9     }
10    for (int i = 1; i < n; i++) {
11        for (int j = 0; j < i; j++) {
12            if (a[j] <= a[i] && dp[j] + 1 > dp[i]) {
13                dp[i] = dp[j] + 1;
14            }
15        }
16    }
17    int max = dp[0];
18    for (int i = 1; i < n; i++) {
19        if (dp[i] > max)
20            max = dp[i];
21    }
22    printf("%d", max);
23 }
```

```
22 |     printf("%d", max);  
23 |     return 0;  
24 | }
```

	Input	Expected	Got	
✓	9 -1 3 4 5 2 2 2 2 3	6	6	✓
✓	7 1 2 2 4 5 7 6	6	6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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